



ESKİŞEHİR TEKNİK ÜNİVERSİTESİ
ESKİŞEHİR TECHNICAL UNIVERSITY

BIM209 – Principles of Software Design and Development

Final Project – Library Book Borrowing System

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Team Members

Özgün Saz - 12125201262

Fatma Bakır - 50782568090

Ece Efe - 20159202742

Vision Statement

The Borrow and Return (B&R) System is a low-cost software application designed to digitalize and simplify the core operations of a small, local library. The motivation behind developing this system is the inefficiencies often seen in manual or outdated library management methods, such as inaccurate inventory counts, difficulties in tracking late returns, and slow service at the counter. The current system relies on paperwork and inconsistent record keeping, which often leads to errors, user dissatisfaction, and excessive administrative time spent by librarians.

The fundamental problem solved by B&R is the modernization and automation of inventory and transaction management. By centralizing the book catalog and all borrowing/return processes, the system provides real-time accuracy regarding the availability of books and eliminates human error in tracking due dates and transaction records.

The system's core features include secure user and librarian authentication, comprehensive catalog search for users, automatic implementation of borrowing limits, and a penalty mechanism for late returns. The system was developed primarily using object-oriented principles in Java, prioritizing usability and reliability.

B&R has two types of users:

1. User: Users benefit from a more efficient and user-friendly experience. They can quickly browse the entire catalog without staff assistance, search for specific books, and instantly determine if a book is available. Clear rules regarding borrowing limits and late return fees are consistently enforced.

2. Librarian: With a reduced manual workload, the librarian can focus on collection development and user assistance rather than paperwork. They gain a robust tool for maintaining book inventory and rely on the system to automatically manage transaction records and apply borrowing policies, thereby increasing operational efficiency.

The successful implementation of this system will provide a more efficient, reliable, and user-friendly library environment, demonstrating the effective application of software design and development principles.

Requirements List

Functional Requirements:

ID	Description
FR1	The system should allow to register as a User role and especially allow users (User or Librarian) to log in using a unique ID/username and password.
FR2	Librarians can add a new book (including ISBN, title, author, genre and quantity) or remove an existing book from the catalog.
FR3	Users and Librarians can view a list of all books in the catalog.
FR4	Users and Librarians can search for books using ISBN or Title Information.
FR5	Users can borrow available books (max 3) or return the book the User successfully borrowed.
FR6	The system must check the User's borrowing limit before allowing a book to be borrowed.
FR7	The system must reduce the available quantity when a book is successfully borrowed or increase its current quantity when a book is successfully returned.
FR8	The system should ban the User from borrowing a new book for one-month if the User returns the book after its Due Date.
FR9	The system must record transaction records (Book ISBN, User ID, Loan Date, Due Date, Return Date).

Non-Functional Requirements:

ID	Description
NFR1	Security: The system should require basic authentications for all users.
NFR2	Usability: The User interface should be simple and understandable.
NFR3	Performance: The system should retrieve the book catalog within seconds.
NFR4	Reliability: The system should not allow the borrowing of books that are out of stock.
NFR5	Implementation: The implementation must include test cases for validating the program's correctness.

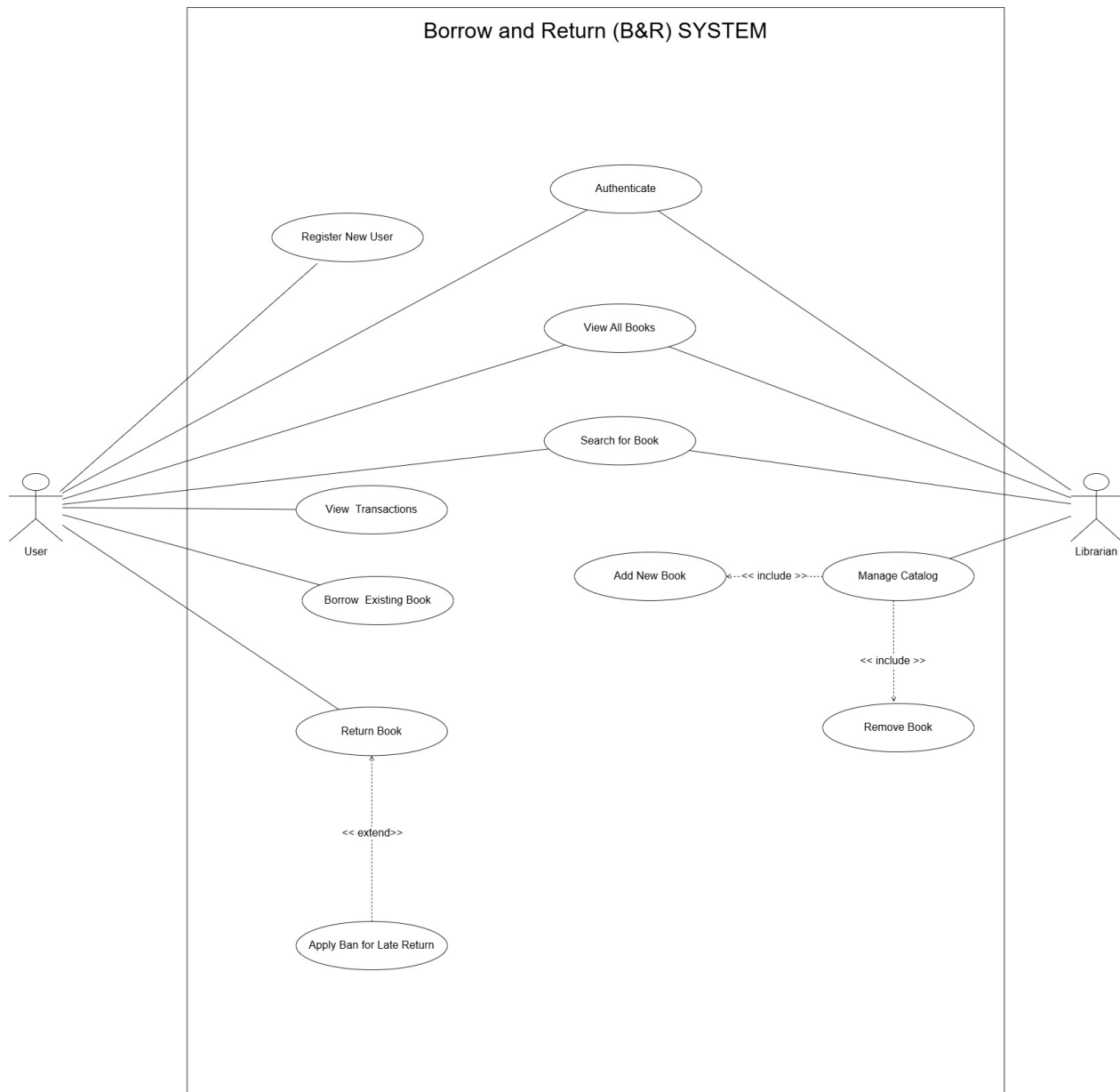
USE CASES

Name	Borrow a Book
Participating actors	User
Entry conditions	The User is successfully logged in (FR1).
Exit conditions	- The book's available quantity decreased (FR7). - A new transaction record is created (FR9).
Flow of events	- The User selects the "Borrow Book" option. - The system shows current borrowing limit and asks for the ISBN of the desired book. - The User enters the ISBN. - The system checks the User's borrowing limit (FR6) and if the User has a borrowing ban. - The system records the transaction records (FR9). - The system displays a confirmation message with the Return Date.
Alternate Step	User is under a borrowing ban (FR8): - The system finds an active one-month borrowing ban. - The system displays an error message: "Borrowing temporarily restricted due to late return. Ban expires on [Date]." - The use case ends.

Name	Return a Book
Participating actors	User
Entry conditions	• The User is successfully logged in (FR1). • The User has a valid transaction record for the book being returned (FR9).
Exit conditions	• The book's available quantity is increased (FR7). • The transaction record is updated with the Return Date (FR9).
Flow of events	1. The User selects the "Return Book" option. 2. User enters the ISBN of the book that they want to return. 3. The system validates the ISBN against transaction records and retrieves the Due Date. 4. The system records the current date as the Return Date (FR9). 5. The system checks if the "Due Date < Return Date" to apply a one-month borrowing ban to the User (FR8). 6. The system updates the transaction records (FR9) and increases the book's quantity (FR7). 7. The system displays a confirmation message, noting that if any penalty is applied.
Alternate Step	Invalid ISBN: 3.1. The entered ISBN does not correspond to a transaction record. 3.2. The system displays an error message: "ISBN is invalid." 3.3. The use case ends.

Name	Manage Book Catalog
Participating actors	Librarian
Entry conditions	The Librarian is successfully logged into the system (FR1).
Exit conditions	The book catalog is updated (either a new book is added, or an existing book is removed) (FR2).
Flow of events	- The Librarian selects the "Manage Catalog" option. - The system displays “Add New Book” or “Remove Existing Book” options. - Depending on the Librarians’ choice, Librarian inputs the book details (title, author, ISBN, quantity, and genre) or chooses which book will be removed from the catalog (FR2). - A new book is added to the system, or an existing book is removed from the system. - The system displays a confirmation message about the operation.

Use Case Diagram



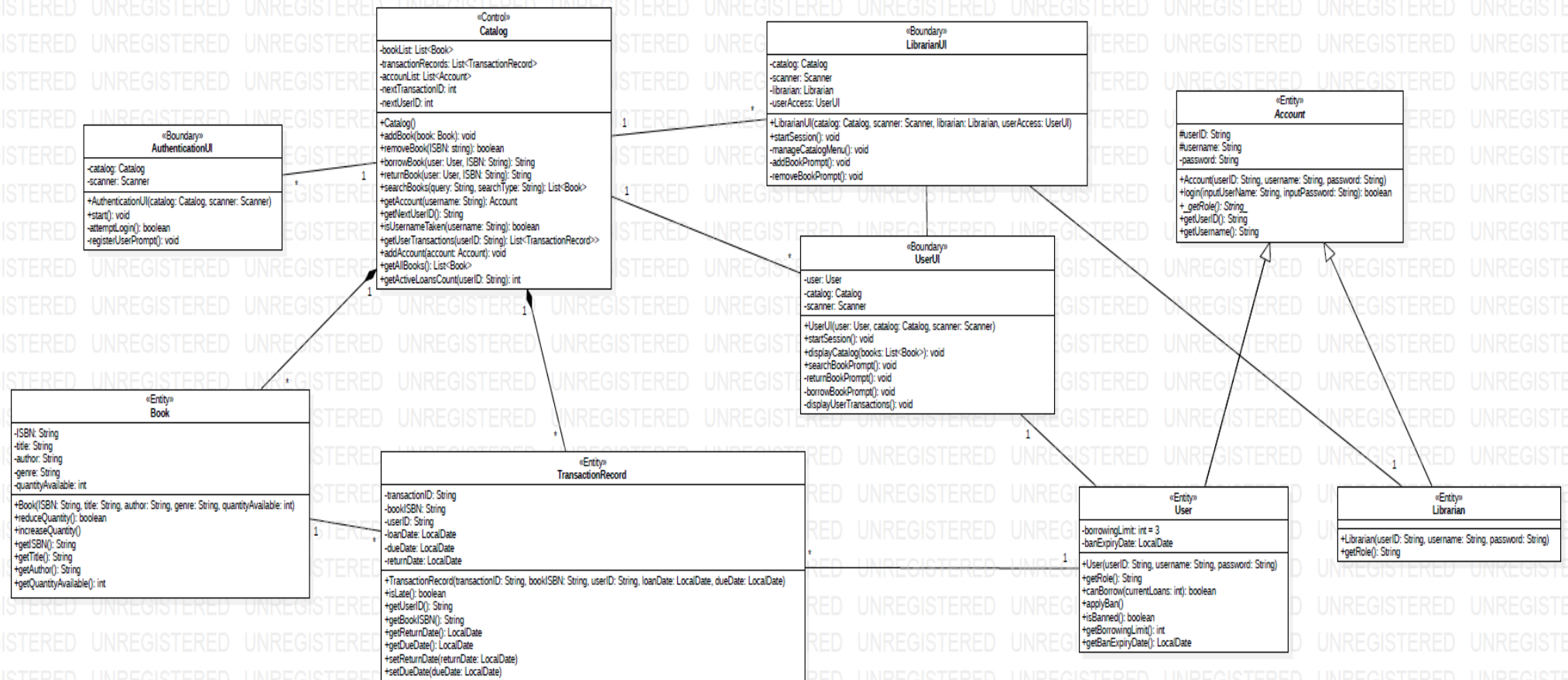
https://drive.google.com/file/d/1_0Q4DGx5zVDPk_6QE8CdwPLWf5fzxbcd/view?usp=sharing

NOUN – VERB ANALYSIS

Noun	Classification	Create a Class?
System	Boundary	-
Account	Entity (Superclass)	+
User	Entity / Actor (Subclass of Account)	+
Librarian	Entity / Actor (Subclass of Account)	+
Book	Entity	+
Catalog	Control	+
Transaction Record	Entity	+
Authentication UI / User UI / Librarian UI	Boundary	+
ID / Username / Password	Attribute	-
Borrowing Limit / Borrowing Ban	Attribute	-
ISBN / Title / Author / Genre	Attribute	-
Quantity	Attribute	-
Loan Date / Due Date / Return Date	Attribute	-
Ban Expiry Date	Attribute	-

Verb	Related Class
log in	User / Librarian
register	User
add / remove	Catalog / Librarian
view / search	Catalog / User
borrow / return	User / Transaction Record
check limit	User
reduce / increase quantity	Book / Catalog
ban	User
record / update	Transaction Record
get / set	All Entity Classes
display	UserUI / LibrarianUI
start / initialize	LibraryApp

UML Class Diagram



https://drive.google.com/file/d/1cSzDYUNjHI9UONL6_970OUoPoJE06F1S/view?usp=sharing

Implementation

The Borrow and Return (B&R) system was implemented using Java, adhering strictly to the object-oriented design principles developed during the analysis phase. The system utilizes a simple console interface for user interaction, which fulfills the Usability requirement by providing a straightforward menu-driven experience. The core of the system is the Boundary-Control-Entity architecture. All persistent data, including Book and Account entities, is centrally managed using Java collections within the Catalog class, ensuring data consistency and centralized control. The design utilizes Inheritance from the abstract Account class to create the concrete User and Librarian roles, which is critical for implementing role-based authentication as required by FR1. This structure maintains high modularity, allowing individual components to be updated without affecting the entire system

```
---- Borrow and Return (B&R) Initializing... ----  
  
---- Welcome to Borrow and Return (B&R) ----  
1. Login  
2. Register New User  
3. Exit  
Enter choice:
```

Figure 1. Login Screen

```
Login Successfull! Welcome, ece (User)  
  
---- User Menu ----  
1. View All Books  
2. Search for Book  
3. Borrow Book  
4. Return Book  
5. View My Transactions  
6. Logout  
Enter choice: 1  
  
---- CATALOG ----  
ISBN | Title | Author | Genre | Availability  
-----  
001 | The Java Handbook | J. Gosling | Tech | Quantity: 3  
002 | Design Patterns Refactored | E. Gamma | Tech | Quantity: 1  
003 | Ah Şu Otizm | Barış Korkmaz | Science | Quantity: 5  
004 | Küçük Kara Balık | Samed Behrengi | Fiction | Quantity: 2
```

Figure 2. User Menu and Viewing the Book Catalog

```
---- YOUR TRANSACTION HISTORY ----  
Transaction ID: 1000 | Book ISBN: 003 | User: U001 | Loan: 2025-12-14 | Due: 2026-01-04 | Returned: 2025-12-14  
Transaction ID: 1001 | Book ISBN: 001 | User: U001 | Loan: 2025-12-14 | Due: 2026-01-04 | Returned: N/A
```

Figure 3. Transaction History After Borrowing and Returning Books

```
Login Successful! Welcome, fatma (Librarian)

---- Librarian Menu ----
1. View All Books
2. Search for Book
3. Manage Book Catalog
4. Logout
Enter choice:
```

Figure 4. *Librarian Menu*

```
---- Manage Catalog Menu ----
1. Add New Book
2. Remove Existing Book
3. Back to Main Librarian Menu
Enter choice:
```

Figure 5. *Manage Catalog option for managing book adding/removing*

```
---- Add New Book ----
Enter ISBN: 2626
Enter Title: Deneme
Enter Author: Ozgun
Enter Genre: Deneme
Enter Quantity to add: 4
Book 'Deneme' added to catalog successfully.
```

Figure 6. *Book Adding Process*

Testing

The core focus of the testing phase was to validate the complete transaction workflows, particularly the complex business logic residing in the Catalog and User classes. This included verifying the integrity of the borrowing limit mechanism (FR5, FR6), the correct application of the ban penalty (FR8), and the management of book availability (FR7, NFR4). A dedicated test script was executed to provide clear, automated validation of these critical rules.

Testing began with a successful borrowing transaction (Test 1). The system correctly processed the loan request, confirming the transaction's success and simultaneously demonstrating compliance with FR7 by reducing the available quantity of the borrowed book to zero. Immediately following this, Test 2 confirmed the system's Reliability (NFR4): an attempt to borrow the now-unavailable book was correctly blocked, returning the message that the book was "not available". This validates the mechanisms designed to prevent over-lending.

The most complex rule, the late return penalty, was validated in Test 3. The test simulated a transaction where the due date was set five days in the past. When the book was returned, the system correctly compared the dates and applied the ban to the User object, resulting in the expected warning message about the one-month restriction. This demonstrated successful implementation of FR8. The final validation (Test 4) confirmed the effectiveness of the ban: a subsequent attempt by the banned user to borrow a different, available book was correctly blocked, with the system returning the message that borrowing was "temporarily restricted".

These results collectively confirm the successful implementation and integration of the core business logic of the B&R system.

```
----- Testing Catalog Business Logic (UC1, UC2, FR5, FR6, FR8) -----

[Test 1] Successful Borrow (FR5):
Result: Success: Book borrowed! Due Date: 2026-01-04
PASS: Borrowing successful and quantity updated.

[Test 2] Borrow Unavailable Book (FR7 Check):
Result: Error: Book is not available or does not exist.
PASS: Correctly blocked borrowing when quantity is 0.

[Test 3] Late Return and Ban (FR8):
Result: Success: Book returned on 2025-12-14. WARNING: Return was late. A one-month borrowing ban has been applied. Ban expires on 2026-01-14.
PASS: Late return resulted in a borrowing ban.

[Test 4] Borrowing while Banned (FR8 Check):
Result: Error: Borrowing temporarily restricted due to late return. Ban expires on 2026-01-14.
PASS: Correctly blocked borrowing while banned.

----- Catalog Test Complete -----
```

Figure 7. *Test Case Results*